

COMPUTER ORGANIZATION PRACTICE SHEET

NIMCET TARGET PAPER • COMPUTER BASICS

ResourceOS

1. In a 4-way set-associative cache memory system with total main memory size of 16 MB, word size of 1 byte, and cache block size of 16 bytes, if the total cache memory size is 64 KB, how many bits are allocated for the Tag field?

- (A) 8 bits
- (B) 10 bits
- (C) 12 bits
- (D) 14 bits

2. A 5-stage instruction pipeline has stage delays of 150 ps, 120 ps, 180 ps, 160 ps, and 140 ps. Interface registers between stages add a delay of 10 ps each. What is the clock cycle time of this pipeline?

- (A) 180 ps
- (B) 190 ps
- (C) 760 ps
- (D) 750 ps

3. Which statement accurately describes the operational behavior of an Exclusive Cache Hierarchy (e.g., L_1 and L_2 caches)?

- (A) A cache line present in L_1 must always be duplicated in L_2
- (B) A cache line resides in at most one level of the cache hierarchy at any given time
- (C) An L_1 miss always triggers a parallel fetch from both L_2 and Main Memory
- (D) Eviction from L_1 automatically invalidates the corresponding entry in L_2

4. In a 16-module low-order interleaved memory system with memory modules numbered 0 to 15, which module handles the byte address 0x004F?

- (A) Module 4
- (B) Module 7
- (C) Module 12
- (D) Module 15

5. A processor uses a 32-bit instruction format containing 3-address fields. If 6 bits are used for the opcode and each register address field specifies one of 32 registers, how many bits remain available for immediate operand values?

- (A) 9 bits
- (B) 11 bits
- (C) 15 bits
- (D) 21 bits

6. Which pipeline hazard occurs when an instruction depends on the result of a previous instruction that has not yet been written back to the register file?

- (A) Control Hazard
- (B) Structural Hazard
- (C) Data Hazard (RAW)
- (D) Branch Misprediction Hazard

7. What is the primary operational advantage of Low-Order Memory Interleaving over High-Order Memory Interleaving?

- (A) Reduces physical address bus size
- (B) Enables concurrent access to consecutive memory addresses
- (C) Eliminates refresh cycles in Dynamic RAM
- (D) Simplifies address decoder hardware complexity

8. Which cache write policy ensures that the main memory is always updated simultaneously with every cache write hit?

- (A) Write-Back
- (B) Write-Through
- (C) Write-Allocate
- (D) Lazy-Write

9. Which protocol is widely utilized in bus-based multiprocessor systems to maintain coherence among multiple local caches?

10. In a microprogrammed control unit, horizontal microinstructions differ from vertical microinstructions in that horizontal microinstructions:

- (A) Use highly encoded control fields with smaller bit lengths
- (B) Allow high degree of parallel control signal generation
- (C) Require decoders for every control field
- (D) Execute sequentially with lower execution speed

11. What is the function of the Program Status Word (PSW) during an external hardware interrupt service cycle?

- (A) Flushes all instruction pipeline queues
- (B) Stores machine status flags and execution state onto stack
- (C) Holds the device vector address of the target controller
- (D) Converts physical memory addresses into logical pages

12. In a vectored interrupt system, how does the CPU determine the starting address of the Interrupt Service Routine (ISR)?

- (A) By polling each I/O device sequentially
- (B) By reading an interrupt vector supplied by the requesting device
- (C) By fetching a fixed hardcoded address from the Program Counter
- (D) By inspecting the opcode field of the Instruction Register

13. What condition causes a Structural Hazard in an instruction pipeline processing architecture?

- (A) Unanticipated conditional branch evaluation
- (B) Hardware resource conflicts between two concurrent stages
- (C) Delayed register writeback operations
- (D) Cache miss during instruction fetch stage

14. Which addressing mode calculates the effective address of an operand by adding a constant displacement value to the Program Counter?

- (A) Base Register Addressing Mode
- (B) Relative Addressing Mode
- (C) Indexed Addressing Mode
- (D) Register Indirect Addressing Mode

15. What is the role of Operand Forwarding (Data Forwarding) in high-performance CPU pipeline designs?

- (A) Resolves control hazards by predicting branch outcomes
- (B) Passes execution results directly to dependent stages without register file writeback
- (C) Eliminates structural conflicts on single-port RAMs
- (D) Reduces memory access latency during page faults

16. In a Fully Associative Cache, how is a requested memory address located inside the cache entries?

- (A) Using a subset index mapped directly to a single set
- (B) Comparing tag bits simultaneously against all cache lines using associative memory
- (C) By calculating hash values from lower order bits
- (D) Sequential search starting from line index zero

17. Which statement regarding Memory Mapped I/O vs. Isolated (I/O Mapped) I/O is correct?

- (A) Memory Mapped I/O uses special distinct CPU instructions for data transfer
- (B) Memory Mapped I/O treats I/O device registers as standard memory locations
- (C) Isolated I/O shares the same address space as main memory
- (D) Memory Mapped I/O requires a separate control line bus for device access

18. What happens when a Dirty Bit is set to '1' in a cache block managed under a Write-Back policy?

- (A) The block contains invalid data and must be fetched again
- (B) The block has been modified and must be written to main memory upon replacement
- (C) The block is currently locked for read-only access
- (D) The block matches main memory contents exactly

19. What distinguishes a CISC architecture from a RISC architecture?

- (A) Fixed length instructions and heavy reliance on registers
- (B) Variable instruction formats with complex addressing modes
- (C) Single cycle execution for all microinstructions
- (D) Exclusive load-store memory access model

20. What is the direct consequence of a Branch Misprediction in a deep execution pipeline?

- (A) Memory refresh cycle interruption
- (B) Pipeline flush and performance penalties from invalid instructions
- (C) Immediate trigger of non-maskable hardware interrupts
- (D) Data corruption in the register file

21. In Daisy Chaining priority interrupt organization, how is interrupt priority determined among requesting peripheral devices?

- (A) By programmable control register priority bits
- (B) By physical electrical location along the interrupt grant line
- (C) By dynamic round-robin scheduling
- (D) By device transfer bandwidth capacity

22. In a CPU architecture using Auto-decrement Addressing Mode, when is the index/register value decremented?

- (A) After accessing the memory operand
- (B) Before accessing the memory operand
- (C) Simultaneously during the execution cycle
- (D) During the instruction decode phase

23. Which cache replacement algorithm replaces the block that has not been referenced for the longest period of time?

- (A) First-In First-Out (FIFO)
- (B) Least Recently Used (LRU)
- (C) Optimal Replacement (OPT)
- (D) Most Frequently Used (MFU)

24. Which bus arbitration technique utilizes a central hardware unit to process bus requests and allocate ownership?

- (A) Distributed Arbitration
- (B) Centralized Arbitration
- (C) Daisy Chain Arbitration
- (D) Static Token Arbitration

25. In memory hierarchy design, spatial locality of reference implies that:

- (A) Items accessed recently are likely to be accessed again soon
- (B) Items stored in nearby memory addresses are likely to be accessed soon
- (C) Execution speed is directly proportional to bus bandwidth
- (D) Cache hits remain constant across loop executions

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(B)	6	(C)	11	(B)	16	(B)	21	(B)
2	(B)	7	(B)	12	(B)	17	(B)	22	(B)
3	(B)	8	(B)	13	(B)	18	(B)	23	(B)
4	(D)	9	(B)	14	(B)	19	(B)	24	(B)
5	(B)	10	(B)	15	(B)	20	(B)	25	(B)